

DIGITAL LOGIC CIRCUIT DESIGN

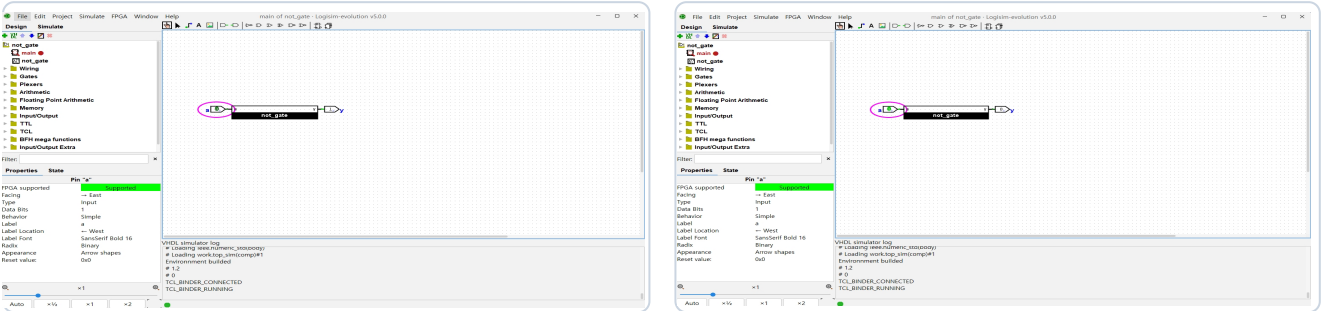
Assignment-1: VHDL Workflow & NOT Gate Co-Simulation

STUDENT	STUDENT ID	DATE	SOFTWARE TOOLS
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1. VHDL SOURCE CODE (NOT_GATE.VHD)

```
library ieee;
use ieee.std_logic_1164.all;
entity not_gate is
  port (
    a : in std_logic;
    y : out std_logic
  );
end entity not_gate;
architecture behavioral of not_gate is
begin
  y <= not a;
end architecture behavioral;
```

2. CIRCUIT DESIGN & CO-SIMULATION RESULTS



3. EXPLANATION OF RESULTS

This experiment verified the basic VHDL workflow and co-simulation environment between **Logisim-evolution** and **Altera Questa Advanced Simulator**.

- **Implementation:** A 1-bit NOT gate entity was declared with input port a and output port y, configured via concurrent signal assignment (`y <= not a;`).
- **Verification:**
 - When input pin a was set to 0, output pin y produced 1.
 - When input pin a was toggled to 1, output pin y produced 0.
- **Conclusion:** The simulation output accurately aligns with the theoretical truth table of a logical NOT gate, confirming successful environment setup and VHDL compilation.